TEXCYCLE AI

COMBINED MODEL EVALUATION REPORT

A Digital Marketplace for Upcycling and Valorizing Sri Lanka's Textile Waste

Research Project: R26-IT-081

Coverage: Member 01–Member 04 model components

Prepared: 19 October 2026

Status: Consolidated draft based on the available final-presentation material

IMPORTANT EVIDENCE NOTE

Metrics in this report are transcribed as previously reported in project presentation material. They are not independently reproduced or certified by this report. Where the evaluation split, target definition, unit, or exact metric is not documented, the result is explicitly marked for verification. No missing score has been invented.

Executive Summary

TexCycle AI is a digital marketplace concept intended to help textile sellers and buyers identify textile materials, understand sustainability indicators, estimate commercial value, discover suitable buyers, and review marketplace trends. This report consolidates the model-related information available for the four member components: textile image analysis, Sustainability Intelligence, Commercial Intelligence, and Marketplace Predictive Analytics.

The available presentation material reports results for image classification, defect analysis, price prediction, buyer recommendation, listing success, sales-time prediction, market-trend classification, and sustainability prediction. However, the material does not consistently specify the held-out test set, dataset size, class distribution, target definition, baseline comparison, or per-model confusion matrix. Therefore, the values below should be treated as reported preliminary results pending verification against training notebooks, evaluation logs, model artifacts, and the relevant test datasets.

Evaluation area	**Model/method**	**Previously reported result**	**Evidence status**
Member 01 – Fabric classification	EfficientNet-B3	Accuracy 89.68%	Verify test split and classes
Member 01 – Defect analysis	MobileNetV2 / U-Net (implementation needs confirmation)	90.87% reported	Exact metric and task unclear
Member 02 – Sustainability	XGBoost regressors + grade classifier	R^2 range 0.9389–0.9887	Map each score to target/model
Member 03 – Price prediction	XGBoost regressor	MAE 12.24; RMSE 15.40; R^2 0.9590	Confirm price unit and test set
Member 03 – Buyer recommendation	Random Forest + XGBoost approach	83.56% reported	Define matching-quality metric
Member 04 – Listing success	XGBoost classifier	Accuracy 83.65%	Define successful-listing label
Member 04 – Sales time	XGBoost regressor	MAE 14.53 days; RMSE 20.87; R^2 0.76	Moderate/limited fit; inspect errors
Member 04 – Market trend	Classification model	Accuracy 98.72%	Check labels, leakage, and test protocol

1. Purpose and Scope

The purpose of this report is to document what each model is intended to do, summarize the results previously reported, explain the appropriate evaluation measures, and identify what evidence is still required before the results can be described as final. The report covers machine-learning models only where model information is available. Traceability workflows and formula-derived environmental indicators are discussed separately because they are not automatically trained machine-learning models.

2. Evaluation Principles

A model must be evaluated using data that were not used to fit the model. Preprocessing steps, encoders, feature selection, and threshold tuning must be fitted using training data only. If the dataset contains generated or synthetic records, that limitation must be disclosed. Very high scores should be checked for data leakage, duplicate records, target leakage, and labels generated directly from the same input features.

Task type	**Recommended metrics**	**How to interpret**
Classification	Accuracy, precision, recall, F1-score, confusion matrix; ROC-AUC where suitable	Accuracy alone can hide poor performance on minority classes.
Regression	MAE, RMSE, R^2; residual plots and baseline comparison	MAE/RMSE must be interpreted in the target's units. R^2 does not show error size by itself.
Image segmentation	IoU, Dice coefficient, precision/recall for defect pixels; plus, clear test examples	A single accuracy percentage may be misleading when defect pixels are rare.
Recommendation/ranking	Precision@K, Recall@K, NDCG@K, hit rate, or a defined expert relevance assessment	A generic percentage is not meaningful until the matching-quality target is defined.
Environmental estimates	Compare against documented factors/baselines and sensitivity analysis	Model estimates and formula-based estimates are not direct measurements of environmental savings.

3. Member 01 – Textile Image Classification and Defect Analysis (IT22281814)

Role in the system: the seller uploads a textile image; the component identifies a fabric category and, where implemented, provides defect-related analysis. The final output should distinguish a predicted class from a segmentation mask or any rule-based condition indicator.

3.1 Fabric classification – EfficientNet-B3

- Purpose: classify an uploaded textile image into one of the supported fabric classes.
- Previously reported result: 89.68% test accuracy.
- Interpretation: if reproduced on an independent and representative test set, this would mean approximately 89.68% of tested images were assigned the correct class. It does not prove equal performance for every class.
- Evidence required: number of images; exact class names and counts; train/validation/test split; image resize and normalization; training configuration; confusion matrix; per-class precision, recall and F1; example correct and incorrect predictions.

3.2 Defect-related model – MobileNetV2 / U-Net

- Purpose: detect defects or identify defect regions, depending on the final implementation.
- Previously reported result: 90.87%; the source presentation does not conclusively establish whether this is classification accuracy, segmentation IoU/Dice, or another metric.
- Model identity must be confirmed: MobileNetV2 is commonly used for image classification or feature extraction; U-Net is a segmentation architecture. They should not be described as interchangeable models.
- Evidence required: exact architecture used in code; whether a defect class or pixel mask is predicted; definition of the 90.87% score; test-set results; per-defect results; and visual examples of masks/predictions.

Evaluation conclusion: fabric classification has a reported numerical result, but reproducibility details are needed. The defect-analysis score should not be presented as a confirmed accuracy result until its metric and model are identified.

4. Member 02 – Sustainability Intelligence (IT22215338)

Purpose: provide users with sustainability-related indicators for textile materials. The component combines trained predictions with formula-based estimates and, where implemented, rule-based grade or reuse-pathway logic.

4.1 Model design

Available project notes describe a fitted preprocessing pipeline using a ColumnTransformer and OneHotEncoder with XGBoost models. The recorded input features are fabric_type, weight_kg, health_score, repairability, condition, district, province, and industry_type.

- Regression outputs recorded in project notes: circularity_score, recyclability_score, carbon_reduction_percent, water_reduction_percent, co2_saved_kg, and water_saved_liters.
- Grade classification: a separate XGBoost classifier predicts a sustainability grade, which is converted to a label using a saved grade encoder.
- Derived indicators mentioned in project notes include energy_saved_kwh, landfill_diverted_kg, sdg_impact_score, and economic_impact_score. These should be labelled formula-derived proxies if they are calculated in code rather than independently predicted.

4.2 Previously reported evaluation

Earlier project notes report R^2 values between 0.9389 and 0.9887 across sustainability experiments. The available summary does not map every value to a particular model, target, dataset version, or held-out split. This range must therefore remain a provisional reported range rather than a single verified model score.

4.3 Required checks and limitations

- Create a results table with one row for each regression target and its own MAE, RMSE, and R^2.
- For sustainability grade classification, report accuracy, precision, recall, F1-score, class distribution, and a confusion matrix.
- Document the environmental baseline, source, units, system boundary, and formula used for CO_2 and water estimates.
- Clearly separate predicted values from formula-derived indicators. For example, a code-defined calculation such as energy_saved_kwh = co2_saved_kg × 1.5 is a proxy based on an assumed factor, not a measured energy saving.

- A previously documented model-loading error reported as an XGBoost 'input stream corrupted' error means that the affected artifact's loading and prediction path must be retested before claiming deployment readiness.

Evaluation conclusion: the component has a clear model-based and formula-based structure, but final evaluation requires target-by-target metrics, reproducible artifact loading, and documented environmental assumptions.

5. Member 03 – Commercial Intelligence (IT22221278)

The presentation identifies Madushanka H.M.T.G. for this component. Its machine-learning focus is indicative price prediction and buyer recommendation. Traceability is a supporting system workflow and should not be reported as an ML model unless a separate trained model is actually used.

5.1 Price prediction – XGBoost Regressor

- Purpose: estimate an indicative textile material price from material, condition, quantity, and market-related features.
- Previously reported result: MAE = 12.24; RMSE = 15.40; R^2 = 0.9590.
- Interpretation: R^2 of 0.9590 suggests that the model explained a high proportion of variance on the evaluation data used for that result. This is not enough to establish real-market accuracy without confirming the test split, target units, and dataset representativeness.
- Evidence required: price target and currency/unit (for example, LKR per kg if that is the actual target); dataset source and size; split strategy; baseline model; MAE/RMSE/R^2 on held-out data; residual analysis; and error across fabric types and price ranges.

5.2 Buyer recommendation – Random Forest + XGBoost approach

- Purpose: rank or recommend potential buyers for a textile listing.
- Previously reported result: 83.56%; the exact metric and evaluation protocol are not defined in the available presentation.
- Evidence required: definition of a relevant match; whether Random Forest and XGBoost are both used in the final implementation; test cases or historical interactions; Precision@K/Recall@K or another justified metric; and comparison with a simple baseline.

Evaluation conclusion: the price model has promising previously reported regression metrics, but the currency/unit and held-out evaluation evidence must be confirmed. The buyer recommendation result needs a defined matching-quality metric before it can be interpreted.

6. Member 04 – Marketplace Predictive Analytics (IT22281678)

The presentation identifies Thalagala T.A.D.V. for this component. The marketplace dataset feature groups include material and condition information, sustainability and price signals, market context, and seller/engagement signals. Project notes also warn that marketplace data include synthetic or prototype records.

6.1 Listing success – XGBoost Classifier

- Purpose: predict whether a listing meets the project's defined success condition.
- Previously reported result: accuracy 83.65%.
- Required definition: specify what success means (for example, sold within a stated time, reached a target number of views, or another label actually used).
- Required metrics: precision, recall, F1-score, confusion matrix, class distribution, and comparison against a majority-class baseline.

6.2 Sales-time prediction – XGBoost Regressor

- Purpose: estimate how many days a listing may take to sell.
- Previously reported results: MAE = 14.53 days; RMSE = 20.87; $R^2 = 0.76$.
- Interpretation: The model has some explanatory value in the reported evaluation, but an average absolute error of 14.53 days may be substantial for sellers. The practical acceptability depends on the normal sales-time range.
- Required checks: compare against a median-days baseline; inspect residuals and large errors; evaluate separately for fabric types, regions and price bands; and confirm that features only use information available at prediction time.

6.3 Market-trend classification

- Purpose: classify the trend as Increasing, Stable, or Decreasing.
- Previously reported result: accuracy 98.72%.
- Critical validation: document how the labels were created, whether the data are time-ordered, and whether future information or the target formula leaked into the input features.
- Required metrics: confusion matrix and per-class precision/recall/F1; chronological holdout testing is preferable where the model predicts future trends.

Evaluation conclusion: listing-success and market-trend results need label and leakage checks. Sales-time prediction should be evaluated for practical error, not R^2 alone. Synthetic/prototype data limit claims about real buyer behaviour, prices, and seasonal demand.

7. Consolidated Results and Verification Register

Component	Task	Reported result	What must be confirmed
Member 01	Fabric class	Accuracy 89.68%	Test split, class labels, per-class metrics
Member 01	Defect analysis	90.87% (metric unclear)	Exact model, task, and metric
Member 02	Sustainability regression	R^2 0.9389–0.9887 range	Map each score to each target and test set
Member 02	Sustainability grade	No separate classification score available	Accuracy, precision/recall/F1 and confusion matrix
Member 03	Price prediction	MAE 12.24; RMSE 15.40; R^2 0.9590	Currency/unit, target, test data, and baseline
Member 03	Buyer recommendation	83.56% (metric unclear)	Matching definition and ranking evaluation
Member 04	Listing success	Accuracy 83.65%	Success label, class balance and test evidence
Member 04	Sales time	MAE 14.53 days; RMSE 20.87; R^2 0.76	Baseline, target range and residual analysis
Member 04	Market trend	Accuracy 98.72%	Label generation, time split and leakage checks

8. Cross-Component Limitations

- Dataset representativeness: synthetic marketplace records may not reflect actual Sri Lankan textile-market prices, buyer preferences, or seasonal demand.
- Reproducibility: the report needs dataset versions, split seeds, training configuration, software versions, saved artifacts, and evaluation scripts.
- Data leakage: all models require checks that target information, future data, duplicates, or label-generation rules have not leaked into the features.
- Metric alignment: classification, regression, segmentation, and ranking require different metrics. A percentage should not be labelled 'accuracy' unless that is what was calculated.

- Environmental claims: CO_2 and water savings depend on documented baselines, units, boundaries, and assumptions. They are estimates and are not independently certified impacts.
- Deployment: a model file existing on disk does not prove that the model loads correctly, returns valid predictions, or is integrated end-to-end in the application.
- System integration: describe automatic data hand-offs only where the current code implements them; otherwise, report the components as connected through shared user workflows.

9. Recommended Final Validation Procedure

1. Freeze and record the exact dataset version for each member's model.
2. Record sample counts, target labels, class distribution, feature list, missing-value handling, and preprocessing steps.
3. Use a held-out test set that was not used for fitting or tuning. Use a chronological split for future-oriented market forecasts where appropriate.
4. Run the final saved model artifact, not only a notebook cell or an in-memory model.
5. Calculate task-appropriate metrics from saved predictions and true test labels.
6. Save evidence: evaluation script, metrics output, confusion matrix or residual plot, representative predictions, and model/version metadata.
7. Check for leakage, duplicates, class imbalance, and baseline performance.
8. Update the results table so every number maps to one exact model, target, dataset, split, and metric.
9. Only then describe a score as a verified final test result.

10. Conclusion

The available project presentation records several promising model results across TexCycle AI's four member components. The strongest defensible conclusion at this stage is that the project has implemented or proposed a multi-model workflow for textile classification, sustainability assessment, price prediction, buyer recommendation, and marketplace forecasting, with preliminary metrics reported for several tasks. The results should not be treated as independently verified until the corresponding test data, metric definitions, evaluation scripts, and saved model artifacts are checked. The highest priority is to resolve the defect-model metric, map sustainability R^2 values to their individual targets, define buyer-match quality, and validate the high market-trend accuracy against leakage.

References and Project Evidence

- Primary project evidence used for this consolidated draft: TexCycle AI final presentation materials, including the Member 01 image-analysis section, Member 02 Sustainability Intelligence section, Member 03 Commercial Intelligence section, Member 04 Marketplace Predictive Analytics section, and the consolidated results/limitations slides.
- Methodological reference named in the project presentation: Tan, M. and Le, Q. V. (2019). “EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks.” ICML.
- Methodological reference named in the project presentation: Sandler, M. et al. (2018). “MobileNetV2: Inverted Residuals and Linear Bottlenecks.” CVPR.
- Environmental context references named in the project presentation: Ellen MacArthur Foundation (2017), A New Textiles Economy: Redesigning Fashion’s Future; United Nations Environment Programme (2023), Sustainability and Circularity in the Textile Value Chain: A Global Roadmap; United Nations, Sustainable Development Goals.